Multimodal Remote Sensing Image Matching Based on Weighted Structure Saliency Feature

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***Abstract*— Matching multimodal remote sensing images (MRSIs) is a challenging task. Due to significant nonlinear radiation differences (NRDs), traditional image-matching methods cannot achieve satisfactory results. This article shows that structural information can get more robust matching results compared with texture information (i.e., gradient features) from images. In order to better explore the structural information of images, this article proposes an MRSI matching method using structure saliency features, called weighted structure saliency feature (WSSF). Two strategies are investigated and integrated into WSSF to improve the matching performance. The scale space is constructed based on the pointwise shape-adaptive texture scale filtering, which can better retain the structure features, and the second-order Gaussian steerable filtering, edge confidence map, and phase features are combined to establish the structural saliency map combined with second-order Gaussian steerable filtering, which is much more robust to NRD than traditional gradient map. The performance of the proposed method was evaluated on a total of 120 image pairs from two MRSI datasets and compared with the state-of-the-art matching methods, including the histogram of the orientation of weighted phase (HOWP), locally normalized image feature transform (LNIFT), co-occurrence filter space matching (CoFSM), radiation-variation insensitive feature transform (RIFT), local phase sharpness orientation (LPSO), and position-scale-orientation scale-invariant feature transform (SIFT) (PSO-SIFT). The experimental results indicate that WSSF obtains satisfactory and reliable results in terms of success rate (SR) and matching accuracy. Compared with the above six methods, the matching accuracy of WSSF is improved by more than 20.275%, and the SR is improved by over 5.833%. The source code will be publicly available at https://github.com/WGY-RS/WSSF.**

***Index Terms*— Log-polar descriptor, multimodal image matching, pointwise shape-adaptive texture filtering (PSATF), structure saliency feature.**

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I. Introduction

IMAGE matching refers to the process of detecting reliable corresponding feature relationships from the same scene images captured in different situations, including sensors, times, and perspectives [1]. Image matching technology is widely used in various fields, such as 3-D reconstruction [2], [3], image fusion [4], [5], change detection [6], [7], [8], [9], and medical image analysis [10]. With the rapid development of multisensor technology, the types of imagery are richer, and the problem of multimodal image matching gradually becomes a hot issue [11]. Different from images captured by a single sensor, multimodal images have obvious temporal, spatial, and nonlinear radiation differences (NRDs) due to different imaging mechanisms, such as visible light images and thermal infrared images. Visible light images capture reflection signals in the visible light band [12], while thermal infrared images capture thermal radiation information from different objects [13]. The difference in imaging mode results in significant NRD between visible light images and thermal infrared images.

Previous research on multimodal image matching generally consists of feature-based, area-based, and deep-learning-based methods [14]. Area-based methods are mainly used for fine matching of remote sensing images [15], [16], [17], [18], [19]. The characteristic of area-based methods is the ability to obtain high matching accuracy. However, these methods perform poorly in the case of image rotation and scale transformation [24]. Feature-based methods mainly use the point, line, and surface features existing in the image to construct descriptors for similarity comparison. These methods are more robust to image geometric transformation. Feature matching methods such as scale-invariant feature transform (SIFT) [20], speeded-up robust features (SURF) [21], local binary pattern (LBP) [22], and synthetic aperture radar (SAR)-SIFT [23] use image gradient features to build descriptors. However, due to the sensitivity of image gradient features to NRD and the large contrast difference between images [24], the above methods are difficult to achieve satisfactory results on multimodal remote sensing images (MRSIs). Compared with image gradient features, the use of phase features brings new breakthroughs [25]. Although phase features are more suitable for MRSI matching, they might have problems with phase extreme value mutations and direction reversal, which can lead to poor invariance to the image rotation [26], [27].

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Deep-learning-based methods have shown great potential in MRSI matching [28], [29], [30], and they are not the best solution in all situations due to the difficulty in collecting adequate and suitable training datasets.

Image information is mainly composed of texture, color, and structure [31]. The texture information corresponding to gradient features is proved to be unable to handle MRSI matching well, while the structure information corresponding to phase features has a better performance [2], [24], [32], [65]. Therefore, we consider exploring the structural information of images for MRSI matching. Most of the existing structural methods rely on the excellent ability of phase congruency (PC) to extract edges, but these methods are easily affected by the issues of phase extreme value mutations and direction reversal. In addition, PC is calculated based on the maximum local energy value; it lacks a comprehensive description of the overall outline of the image. Instead of using phase features, Wu et al. [33] employed visual landmarks and proposed a descriptor rearrangement (DR) matching strategy. This method has a good tolerance to rotation and scale changes, but it takes a long time to run. Ye et al. [42] proposed to adopt LSS descriptors with normalization by the L2 norm, which can achieve better intensity invariance. However, as this method is free of the calculation of the main orientation for descriptors, it cannot solve the problem of rotation. Yao et al. [11] proposed a method using co-occurrence filtering (COF) [34] to smooth images and preserve structure features. In most cases, this method is scale- and rotation-invariant. However, COF can cause halo artifacts and blurring around the edges of the image, which will bring difficulty to the matching of MRSI with a strong structure.

To fully explore structure features for MRSI matching, a novel MRSI matching method is proposed in this article, called weighted structure saliency feature (WSSF). First, pointwise shape-adaptive texture filtering (PSATF) is adopted to smooth images and preserve structure features. Second, the second-order Gaussian steerable filtering, edge confidence map (ECM), and phase features are weighted to establish the structural saliency map combined with the second-order Gaussian steerable filtering ($G^{\theta}_{2,\sigma}$ − SSM). ECM mainly reflects the contour features of image, and phase features collect local structural information. Thus, $G^{\theta}_{2,\sigma}$ −SSM is more suitable for MRSI description than traditional gradient map. The contributions of this article are summarized as follows.

1) A pointwise shape-adaptive texture scale space is designed, which can smooth images and preserve structure features well. Meanwhile, PSATF solves the problems of glitches and blurring in densely structured areas in previous studies [11].
2) The $G^{\theta}_{2,\sigma}$−SSM is proposed, which is different from other methods reliant on phase features and mainly depends on contour features (ECM). $G^{\theta}_{2,\sigma}$−SSM is more robust to NRD than traditional gradient maps and can deal with issues such as phase extreme value mutations and direction reversal better than phase features can.
3) A robust MRSI matching method called WSSF is developed, which achieves superior performance in terms of the number of correct matches (NCMs), success rate (SR), and matching accuracy.

II. Related Work

MRSI matching methods are mainly divided into area-based, feature-based, and deep learning-based methods, which will be briefly reviewed in the following.

A. Area-Based Methods

Area-based methods mainly use matching templates around predefined feature points (which can be preset using geographic coordinate points [35]) to perform high-precision matching, and sometimes, the entire image is used as a matching template. The keypoint is to establish a reasonable and effective similarity measure. Early template matching methods used the grayscale and intensity features of images to find corresponding points, such as mutual information (MI) [36], and normalized cross correlation (NCC) [37]. NCC measures the linear change relationship between matching templates, which is very sensitive to the nonlinear gray level changes, and has been proven that it is difficult to be used to match MRSI [38]. Although MI can deal with the NRD, it is time-consuming [39] and is easy to fall into the local optimal solution. The subsequent methods mainly use the phase information of images [15], [16], [17], [18], [19], [40], [41]. Unlike the gradient feature, the phase feature can reflect the structural changes of images, thus avoiding the matching failure caused by different gray changes in the region. Ye et al. [42] used phase orientation and amplitude to build descriptors and combined with NCC [37] to capture structure features of images, achieving superior matching performance. However, this method is still very sensitive to rotation and scale changes.

B. Feature-Based Methods

Feature-based methods extract common salient features from the images to be matched, such as points, lines, and regions. This section will focus on the methods based on point features, which mainly include three parts: feature detection, such as histogram of the orientation of weighted phase (HOWP) [24], Harris [43], difference of Gaussian (DoG) [44], and nonlinear diffusion-based Harris–Laplace (NDHL) [45], feature description, such as SIFT, SURF, oriented FAST and rotated BRIEF (ORB) [46], radiation-variation insensitive feature transform (RIFT) [25], and hybrid-domain features and adaptive description (HFAD) [64], and correspondence estimation, such as fast sample consensus (FSC) [47], grid-based motion statistics (GMS) [48], and vector field consensus (VFC) [49].

The matching of homologous images has been well-developed in the past, such as SIFT, SURF, and ORB. These methods mostly use traditional gradient features of images to obtain robust feature descriptors. However, these methods perform poorly on MRSI matching due to significant NRDs between images. To solve this problem, some improvements have been presented, such as SAR-SIFT and

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position-scale-orientation SIFT (PSO-SIFT) [50]. These methods have achieved good results on certain types of MRSI, but are not satisfactory on others, which limits their application to a large extent.

Considering the limitations of traditional gradient features (texture) in MRSI matching, structure features for MRSI matching have been adopted. Wu et al. [33] used visual landmarks and proposed a DR registration strategy. Zhao et al. [51] generated feature descriptors by obtaining the edge segments and combining the corner points. This method solves the image rotation problem but does not consider the scale difference. Li et al. [25] proposed to use the maximum index map (MIM) to deal with the MRSI matching. This method introduces the phase feature to replace the traditional gradient feature, which shows more robustness to NRDs. As phase features can capture image edges and structures, using phase features for MRSI matching has become the mainstream. For example, Fan et al. [52] used a two-step matching method to perform the MRSI matching from coarse to fine; Yao et al. [26] investigated the weight phase orientation to solve the rotation problem of MRSI matching. However, there are still limitations for phase features due to phase extreme value mutation and direction reversal. Recently, Yao et al. [11] adopted COF to obtain the structure features and used the second-order gradient feature to reduce the impact of NRD, which provided a new idea for MRSI matching.

C. Deep-Learning-Based Methods

In recent years, deep learning has been increasingly used in MRSI matching [28], [29], [30], [53], [54], [55], [56]. Wang et al. [53] first paired patches from images and then directly learned the mapping between the patch pairs and the matched labels for MRSI matching. Ye et al. [56] used a deep convolutional neural network (CNN) to calculate deep features, then combined the extracted features with SIFT to obtain combined features, and integrated the combined features into PSO-SIFT [50] for matching. Efe et al. [28] designed a network for multimodal image feature learning and proposed a self-extracting feature learning method to enhance deep network optimization with more similarity information. The deep learning method has the advantages of fast and automatic optimization of parameters and construction of required descriptors [63]. However, deep learning methods rely on a large amount of data (supervised learning-based methods) [29], [53] or time (unsupervised learning methods) [62] for training, and the model effect probably is poor transferability, resulting in a decline in the effect of different test datasets [59]. Thus, further research is needed to achieve more vital generalization ability and better applicability.

III. Methodology

Fig. 1 shows the framework of the proposed WSSF method. The scale space is first constructed using the PSATF. Then, the $G^{\theta}_{2,\sigma}-$SSM is generated according to the phase information, ECM, and second-order Gaussian steerable filtering ($G^{\theta}_{2,\sigma}$). Features from accelerated segment test (FAST) detector [60] is employed to detect keypoints on $G^{\theta}_{2,\sigma}-$SSM, and then, these keypoints are separately described using the improved gradient location and orientation histogram (GLOH) descriptor [61]. Finally, the correspondence is established by using the FSC method [47].

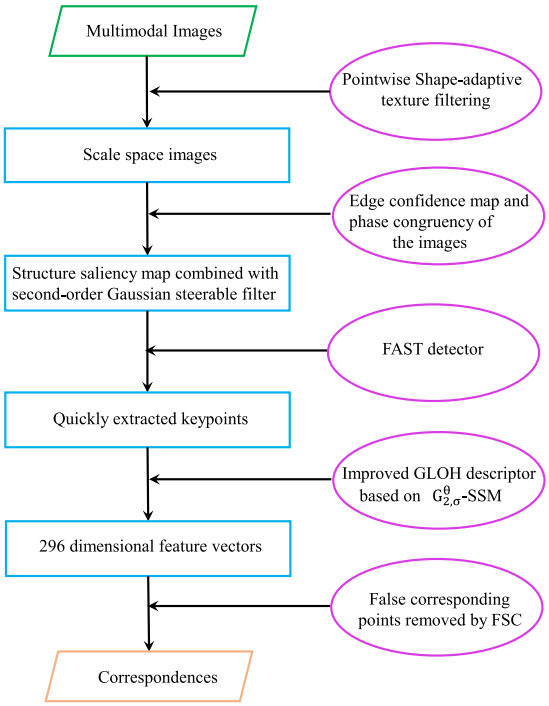


Fig. 1. Framework of the proposed WSSF method.

A. Pointwise Shape-Adaptive Texture Scale Space

The edge-preserving filtering can smooth image and highlight the structure saliency feature. Although the images obtained by different sensors have various texture representations, the structural features are basically the same. At the same time, the image noise can also be suppressed. In order to effectively reduce the image noise and preserve the edges, we propose to construct the scale space using the PSATF, which is an edge-preserving filtering [57].

1) PSATF: PSATF is implemented in the framework of joint bilateral filtering. The definition of PSATF is shown as

$$J_p = \frac{\sum_{q\in N(p)} G_{\sigma_s}(p,q)\cdot M_{\sigma_r}\left(I_p, I_q\right)\cdot I_q}{\sum_{q\in N(p)} G_{\sigma_s}(p,q)\cdot M_{\sigma_r}\left(I_p, I_q\right)} \tag{1}$$

where J_p and I_q are the pixel values of the image output and input; p and q are the index positions of pixels in the image; $G_{\sigma_s}(p,q)\cdot M_{\sigma_r}\left(I_p, I_q\right)$ denotes the weight of pixel q to the output of pixel p; $G_{\sigma_s}(p,q)$ is the Gaussian filtering (GF) of scale σ_s; and $M_{\sigma_r}\left(I_p, I_q\right)$ is a filtering that guides the image with scale σ_r and σ_r is fixed as 0.05.

2) Pointwise Shape-Adaptive Texture-Guided Images: The calculation of the guide image determines the final effect of filtering. In order to obtain the guide image, the texture cleanliness at each point is calculated first. For each point p as the center, a line segment of length k is formed in n directions. To find the direction that is least likely to pass through the significant structure, the ECM [58] based on machine learning

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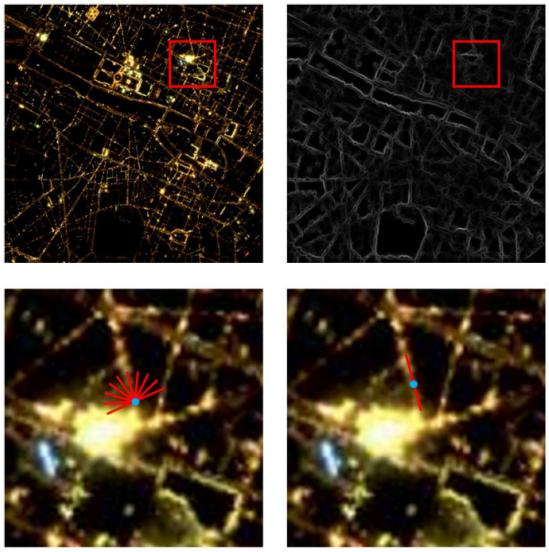


Fig. 2. "Cleanliness" measure adopted in this study.

is used to calculate the edge confidence value (ECV) of each directional line segment, so as to achieve the cleanest line segment $L_\theta(p)$, as shown in Fig. 2. The definition of $L_\theta(p)$ is expressed as

$$\mathrm{CL}(l_\theta(p)) = \mathrm{Var}(\mathrm{ECV}(l_\theta(p))) \tag{2}$$

$$L_\theta(p) = \underset{l_\theta(p)}{\operatorname{argmin}}\{\mathrm{CL}(l_\theta(p))|l_\theta(p)\} \tag{3}$$

where $l_\theta(p)$ represents a line segment with p as the center, direction θ, and length k; $\mathrm{ECV}(l_\theta(p))$ represents the structural confidence of pixels on line segment $l_\theta(p)$. Var is the variance of line segment $\mathrm{ECV}(l_\theta(p))$. $\mathrm{CL}(l_\theta(p))$ represents the cleanliness of the line segment; and $L_\theta(p)$ is the line segment least likely to cross the edge structure.

After getting the texture cleanliness at each point, we calculate the guide image, which is defined as

$$\begin{cases} G(p) = \dfrac{1}{W(p)} \displaystyle\sum_{\theta=\{o_1,\ldots,o_{no}\}} W_\theta(p) B_\theta(p) \\ W_\theta(p) = \exp(-\mathrm{CL}(l_\theta(p))) \\ B_\theta(p) = \dfrac{1}{k} \displaystyle\sum_{q\in L_\theta(p)} I(q) \end{cases} \tag{4}$$

where $G(p)$ represents the guide image; $W_\theta(p)$ is used to control the degree of image smoothing, the higher the value of $W_\theta(p)$, the output will be smoother, $W(p) = \sum_{\theta=\{o_1,\ldots,o_{no}\}} W_\theta(p)$; $B_\theta(p)$ represents the average pixel value of the line segment $L_\theta(p)$; and $I(q)$ is the pixel value of the image I at point p.

3) PSATF Scale Space: The PSATF scale space consists of N layers. The input of each layer is the filtering output of the previous layer, and the filtering is calculated through (1)–(4). The scale of each layer is defined as

$$\sigma_{s_n} = \sigma_{s_0} \cdot \sqrt[3]{2^n}, \quad (n = 0, 1, \ldots, N) \tag{5}$$

where σ_{s_0} represents the scale of the GF of the first layer in scale space and σ_{s_n} corresponds to σ_s in (1); and n represents the nth layer in scale space.

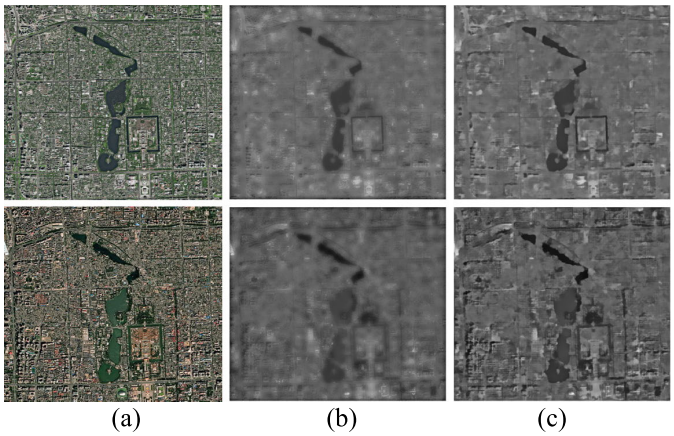


Fig. 3. Filtering results of COF and PSATF. (a) Original image pairs. (b) Results of COF. (c) Results of PSATF.

The filtering window size of each layer is calculated by the initial filtering window and the scale of each layer is as

$$\begin{cases} \mathrm{OC}_n = \dfrac{{\sigma_{S_n}}^2 \cdot \mathrm{NO}}{2} \\ \mathrm{PSATFSpace} = \left\{\mathrm{OC}_n \cdot J_p^n\right\}_{n=0}^{N} \end{cases} \tag{6}$$

where PSATFSpace denotes the image set of PSATF scale space; OC_n is the window size of the nth layer; NO is the initial filtering window size of PSATF (set to 5 in this article); and J_p^n represents the nth filtered layer in scale space.

The filtering results of PSATF and COF are shown in Fig. 3. It can be found that the results of PSATF are better than COF. COF possibly leads to blurring, halo artifacts, and other problems in images, while PSATF can smooth images and highlight structure features well. The PSATF scale space will be used for constructing $G_{2,\sigma}^{\theta}-$SSM and descriptors in Sections III-B and III-C.

B. Structure Saliency Map Combined With Second-Order Gaussian Steerable Filtering ($G_{2,\sigma}^{\theta}-SSM$)

Existing MRSI matching methods mostly use phase orientation to construct feature descriptors. However, due to problems such as NRDs and direction reversal among MRSI, this method often fails to find the correct corresponding points in the matching process [18], [26], [27]. RIFT uses the MIM instead of phase orientation to build feature descriptors, which achieves good results. Inspired by RIFT, we propose to compute feature descriptors using the structural salience of images called $G_{2,\sigma}^{\theta}-$SSM. The calculation of $G_{2,\sigma}^{\theta}-$SSM can be divided into three steps: 1) calculating the ECM of the image, which is described in Section III-A2; 2) computing the maximum moment map ($M_{\max}$) and minimum moment map ($M_{\min}$) according to the phase information; and 3) weighting ECM, $M_{\max}$, $M_{\min}$, and $G_{2,\sigma}^{\theta}$ to obtain the $G_{2,\sigma}^{\theta}-$SSM. Fig. 4 illustrates the construction of $G_{2,\sigma}^{\theta}-$SSM.

1) Maximum Moment Map and Minimum Moment Map: The $M_{\max}$ shows the orientation information of image features, while the $M_{\min}$ reflects the significance of image features. To achieve the $M_{\max}$ and $M_{\min}$, we first calculate the PC on

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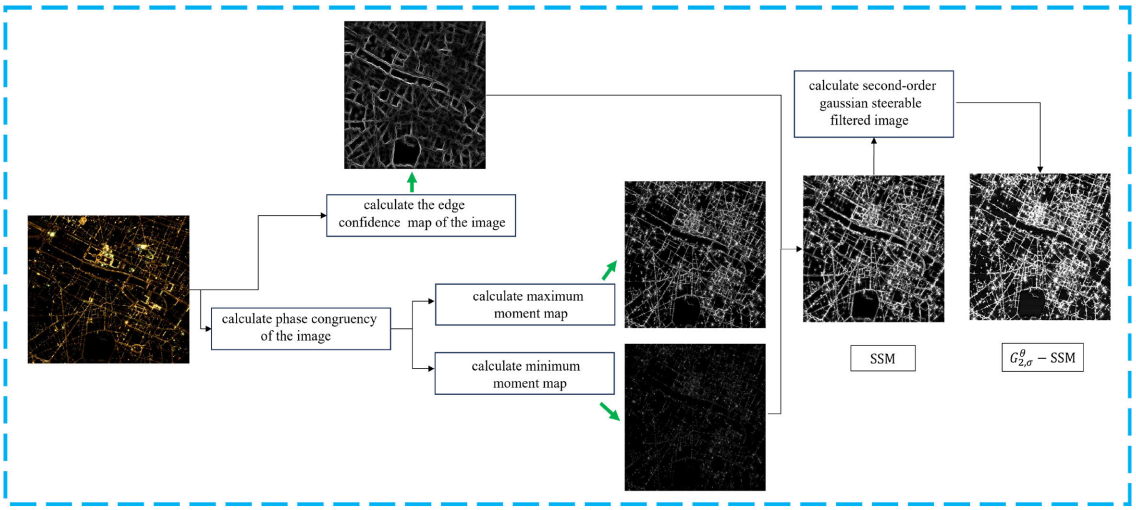


Fig. 4. Construction of $G^{\theta}_{2,\sigma}$ − SSM.

each layer of PSATF scale space as [18]

$$\mathrm{PC}(x,y)=\frac{\sum_o\sum_n W_o(x,y)\lfloor A_{no}(x,y)\Delta\Phi_{no}(x,y)-T\rfloor}{\sum_o\sum_n A_{no}(x,y)+\varepsilon} \tag{7}$$

where $\mathrm{PC}(x,y)$ represents the magnitude of the local PC; (x,y) represents the point coordinate of an image point; $A_{no}(x,y)$ represents the amplitude of wavelet at scale n and direction o at (x,y); and $\Delta\Phi_{no}(x,y)$ is a phase deviation. $W_o(x,y)$ is the weight factor for frequency expansion; T represents the noise threshold; ε is a small constant to avoid having the denominator 0 in the calculation; and $\lfloor\rfloor$ denotes if the value of the closed quantity is more than 0, its value equals itself; otherwise, it is 0.

Subsequently, the $M_{\max}$ and $M_{\min}$ are calculated as

$$\begin{cases} M_{\max}=\dfrac{1}{2}\left(c+a+\sqrt{b^2+(a-c)^2}\right) \\ M_{\min}=\dfrac{1}{2}\left(c+a-\sqrt{b^2+(a-c)^2}\right) \\ a=\displaystyle\sum_{\theta}(\mathrm{pc}(\theta)\cos(\theta))^2 \\ b=2\displaystyle\sum_{\theta}(\mathrm{pc}(\theta)\cos(\theta))(\mathrm{pc}(\theta)\sin(\theta)) \\ c=\displaystyle\sum_{\theta}(\mathrm{pc}(\theta)\sin(\theta))^2 \end{cases} \tag{8}$$

where $M_{\max}$ is the maximum moment map; $M_{\min}$ is the minimum moment map; pc is the magnitude of local PC; and θ is the orientation of the PC.

2) Construction of $G^{\theta}_{2,,\sigma}-SSM$: The calculation of SSM is obtained by weighting ECM, $M_{\max}$, and $M_{\min}$ obtained in steps (1) and (2) as

$$\mathrm{SSM}=k_1*M_{\max}+k_2*M_{\min}+k_3*\mathrm{ECM} \tag{9}$$

where k_1, k_2, and k_3 are the weight coefficients used to measure the proportion of $M_{\max}$, $M_{\min}$, and ECM. Since SSM mainly collects the overall structural information of image, it lacks a description of the local spatial information. In order to better capture distinctive features of different multimodal images, steerable filtering is applied to enhance local spatial information, which can provide more details by collecting more information from different angles [35]. Especially, the second-order Gaussian steerable filtering is used to further process SSM and generate the final $G^{\theta}_{2,\sigma}-$SSM, as it has been proven to be able to effectively deal with NRD and other problems in MRSI matching [35]. The second-order Gaussian steerable filtering is defined as

$$\begin{cases} G^{0^\circ}_{2,\sigma}=G_{xx}=\left(-\dfrac{1}{2\pi\sigma^4}\right)\left(1-\dfrac{x^2}{\sigma^2}\right)e^{\frac{-\left(x^2+y^2\right)}{2\sigma^2}} \\ G^{90^\circ}_{2,\sigma}=G_{yy}=\left(-\dfrac{1}{2\pi\sigma^4}\right)\left(1-\dfrac{y^2}{\sigma^2}\right)e^{\frac{-\left(x^2+y^2\right)}{2\sigma^2}} \\ G_{xy}=\dfrac{xy}{2\pi\sigma^6}e^{\frac{-\left(x^2+y^2\right)}{2\sigma^2}},\ G^{60^\circ}_{2,\sigma}=G_{yy}-G_{xy},\ G^{120^\circ}_{2,\sigma}=G_{yy}+G_{xy} \\ G^{\theta}_{2,\sigma}=\cos^2(\theta)G^{0^\circ}_{2,\sigma}+\sin^2(\theta)G^{60^\circ}_{2,\sigma}-2\cos(\theta)\sin(\theta)G^{120^\circ}_{2,\sigma} \end{cases} \tag{10}$$

where $G^{\theta}_{2,\sigma}$ can be derived from the combination of $G^{0^\circ}_{2,\sigma}$, $G^{60^\circ}_{2,\sigma}$, and $G^{120^\circ}_{2,\sigma}$, and σ represents the scale of filters. The convolution of SSM with $G^{\theta}_{2,\sigma}$ is summed over six directions $(0,(\pi/6),(2\pi/6),(3\pi/6),(4\pi/6),(5\pi/6))$ to obtain the final $G^{\theta}_{2,\sigma}-$SSM.

C. Feature Point Extraction and Log-Polar Coordinate Descriptor

1) Feature Point Extraction: After constructing the scale space, it is necessary to extract the feature points. We use the FAST detector to obtain feature points on the image. On the $G^{\theta}_{2,\sigma}-$SSM of each layer, 5000 feature points are selected according to the response intensity, and a total of 15 000 feature points are selected (a total of three layers of scale space).

2) Improved Log-Polar Coordinate Descriptor Based on $G^{\theta}_{2,,\sigma}-SSM$: Robust descriptors are important for MRSI matching. Since traditional gradient features are very sensitive to image noise, we use $G^{\theta}_{2,\sigma}-$SSM to extract structure gradient features for descriptor construction. The calculation process is

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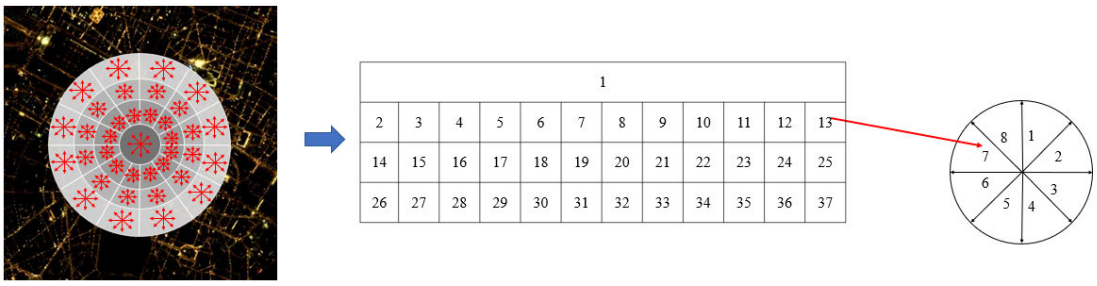


Fig. 5. Log-polar coordinate descriptor for MRSI grid optimization.

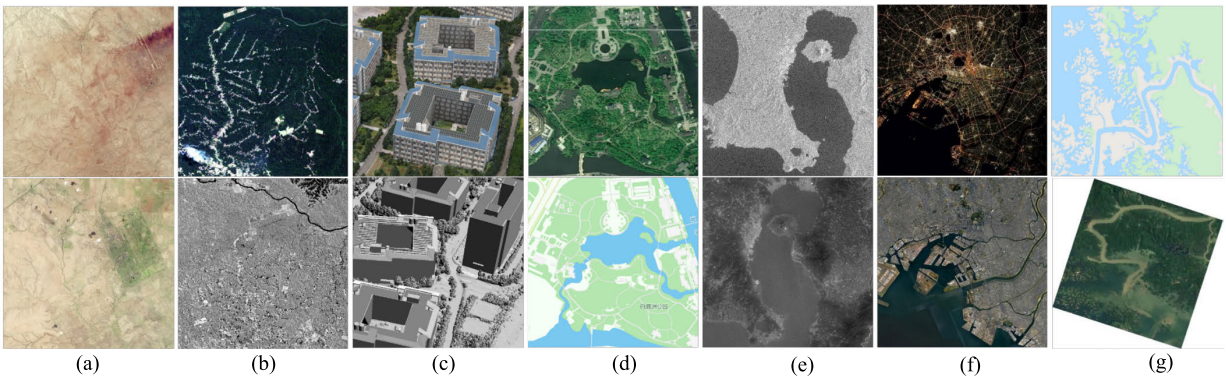


Fig. 6. Sample data of MRSI dataset 1. (a) Optical-optical. (b) Optical-infrared. (c) Optical-depth. (d) Optical-map. (e) Optical-SAR. (f) Day-night. (g) Rotation.

shown as

$$\Gamma_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 0 \end{bmatrix}, \quad \Gamma_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \tag{11}$$

$$\begin{cases} G = \sqrt{(I(x,y) * \Gamma_x)^2 + \left(I(x,y) * \Gamma_y\right)^2} \\ \text{Angle} = \arctan\left(\dfrac{I(x,y) * \Gamma_y}{I(x,y) * \Gamma_x}\right) \end{cases} \tag{12}$$

where Γ_x and Γ_y represent the Sobel operator in the direction of x and y; $I(x, y)$ is the input image; G represents the amplitude of the gradient map; $*$ denotes the convolution operation; and Angle is the gradient orientation.

The GLOH, which is extended from the SIFT descriptor and has good stability, is used to construct feature vectors. First, a local block with the pixel size of $R \times R$ is selected around each feature point, and the main orientation of the local block is calculated. The direction histogram of the local block is evenly divided into 24 equal parts (bin) at 15° intervals, calculating the sum of the structural gradient amplitudes of each bin, and the bin with the highest score is selected as the main orientation. Furthermore, the bin scores greater than 0.9 times in the main orientation will also be recorded as the auxiliary orientation [20]. After obtaining the main orientation of each local block, the whole local image block is rotated according to the main orientation. The GLOH descriptor divides the local feature block into two annular regions and a central circle. The orientation of all pixels in the circular area is summarized over eight directions; each annular area consists of 16 fan areas, and the orientation of all pixels in each fan area is also divided into eight directions. Therefore, the GLOH descriptor will generate an eigenvector of $(2 * 16 + 1) * 8 = 272$ dimensions for each feature point. To improve the stability, we divide the local feature block into three annular regions and a central circle; each annular region consists of 12 sector regions. Finally, a vector of $(3 * 12 + 1) * 8 = 296$ dimensions is generated, as shown in Fig. 5. This not only reduces the description instability of GLOH feature descriptors due to fewer grids, but also does not largely increase the computational overhead.

IV. Experimental Results and Analysis

A. Datasets

In this article, two MRSI datasets are collected for matching evaluation. The details of each dataset are described in the following. Figs. 6 and 7 show several sample image pairs from the collected datasets.

Dataset 1 is collected from Yao et al. [11], which includes a total of six types of MRSI (optical-optical, optical-SAR, optical-depth, optical-map, day-night, and optical-infrared), and each type contains ten image pairs. The images mainly feature the NRD and translational transformation. For each image pair, the provider manually selects about 10–30 corresponding feature points, which are used to calculate the true transformation matrix. In order to better verify the performance of the proposed method on rotation changes, ten image pairs are randomly selected from the dataset and perform rotation transformations. Ultimately, the dataset comprises 7×10 image pairs.

Dataset 2 is collected from Zhang et al. [24], which includes five types of MRSI (optical-SAR, optical-depth, day-night, optical-map, and optical-infrared), and each type contains ten image pairs. Compared to Dataset 1, Dataset 2 is more

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Fig. 7. Sample data of MRSI dataset 2. (a) Optical-depth. (b) Optical-infrared. (c) Optical-depth. (d) Optical-SAR. (e) Day-night.

challenging. All images not only have NRD and contrast differences, but also have geometric variations, including translation, scale, and rotation changes. The true transformation matrix for each image pair is provided, and the image sizes are all between 381 and 750 pixels.

B. Experimental Settings

To validate the effectiveness of the proposed WSSF method, we compare it with six recent advanced methods: HOWP (2023), locally normalized image feature transform (LNIFT) (2022) [1], co-occurrence filter space matching (CoFSM) (2022) [11], RIFT (2019), local phase sharpness orientation (LPSO) (2022) [27], and PSO-SIFT (2017) [50]. The implementation of the above methods is obtained from the author's website. The critical parameters of all comparison methods are set to be the same. Specifically, we set the neighborhood radius of all comparison methods to 48 and the scale difference of images to 1.6. The number of keypoints in each layer of all methods is set to 5000, and the max number of layers is set to 3. The FSC method is used for mismatch elimination for all methods. In other words, we build scale space for LNIFT and ignore the step of optimization of MRSI position in CoFSM.

The NCMs, the ratio of corrected matches (RCMs), the SR, and root-mean-square error (RMSE) are used as quantitative evaluation metrics. NCM represents the number of correct matches, and RMSE represents the root pixel error between corresponding feature points after the true matrix transformation T_{truth}. The definition of correct match is shown in (13), and the calculation process of RMSE is shown in (14), which reflects the accuracy of matching. If the value of RMSE for one image pair is larger than three pixels, it is considered as a failed matching trial

$$\text{correct match} = |\text{ref}(x, y) - T_{\text{truth}} * \text{sen}(x, y)| < 3 \quad (13)$$

where $\text{ref}(x, y)$ represents a point on the reference image; $\text{sen}(x, y)$ is the corresponding feature point of $\text{ref}(x, y)$ on the sensed image; and T_{truth} is the true transformation matrix

$$\text{RMSE} = \sqrt{\frac{1}{\text{NCM}} \sum_{i=1}^{\text{NCM}} \left[\left(x_i - x_i'\right)^2 + \left(y_i - y_i'\right)^2 \right]} \quad (14)$$

where (x_i, y_i) is a point of correct matches; and (x_i', x_i') is the corresponding feature point of (x_i, y_i) after the true matrix transformation.

SR is the ratio of the successful matching trials to the number of all matching trials, and RCM represents the ratio of corrected matches among all matches, whose definition is expressed as

$$\text{RCM} = \frac{\text{NCM}}{\text{NCM} + \text{NFM}} \quad (15)$$

where NFM is the number of false matches.

C. Parameter Analysis

The proposed WSSF mainly includes three parameters, N, R, and R_f. The parameters that need to be analyzed are listed in Table I, and the other parameters are set according to the reference study.

N represents the number of layers in the scale space. In general, the larger the value of N, the more the feature points will be extracted. However, it will also greatly increase the computational complexity of the algorithm. R represents the radius of the feature descriptor, and increasing in R will help the feature descriptor contain more spatial information. If R is too small, the descriptor will not fully reflect the uniqueness, resulting in matching failure. R_f represents the size of the initial PSATF filtering window. The size of R_f decides the smooth degree of the filtered image. The larger the R_f is, the filtered image will be smoother, and the fewer feature points will be captured. On the contrary, the more feature points will be extracted. In addition, R_f also decides the robustness of the filtered image to NRD.

In order to achieve the appropriate parameters, we randomly selected 12 image pairs from two datasets, and the selected image pairs covered all image types. We design three experiments to explore the parameter settings of N, R, and R_f. Only one parameter acts as a variable in each experiment, and the others are fixed. NCM and SR are used as evaluation metrics. Tables II–IV show the results of three parameter experiments. According to Tables II–IV, the following can be inferred.

1) If the parameter N is too small, SR and NCM will be poorer. When parameter N increases, SR and NCM will also be improved, as shown in Table II. However, too large N will significantly increase computational

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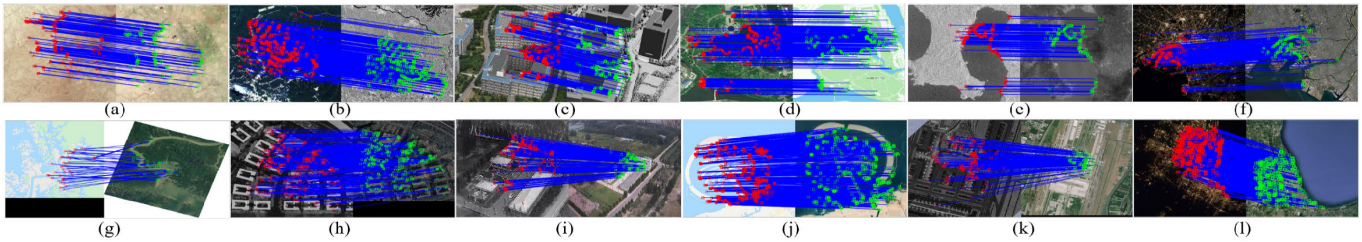


Fig. 8. Matching results of WSSF on sample images. (a) Optical–optical. (b) Optical–infrared. (c) Optical–depth. (d) Optical–map. (e) Optical–SAR. (f) Day–night. (g) Rotation. (h) Optical–depth2. (i) Optical–infrared2. (j) Optical–map2. (k) Optical–SAR2. (l) Day–night2.

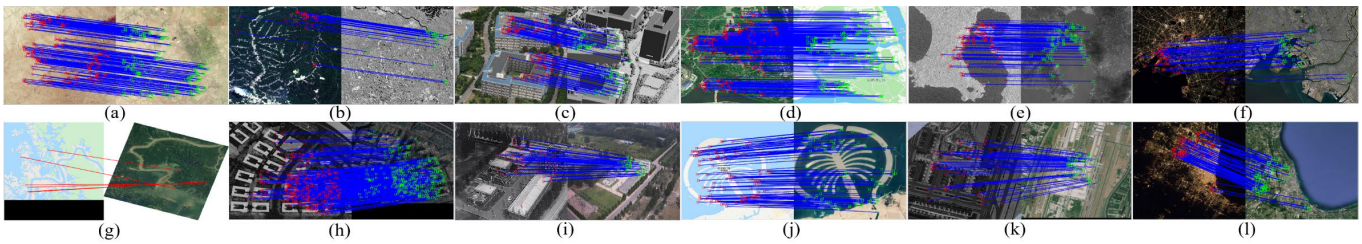


Fig. 9. Matching results of HOWP on sample images. (a) Optical–optical. (b) Optical–infrared. (c) Optical–depth. (d) Optical–map. (e) Optical–SAR. (f) Day–night. (g) Rotation. (h) Optical–depth2. (i) Optical–infrared2. (j) Optical–map2. (k) Optical–SAR2. (l) Day–night2.

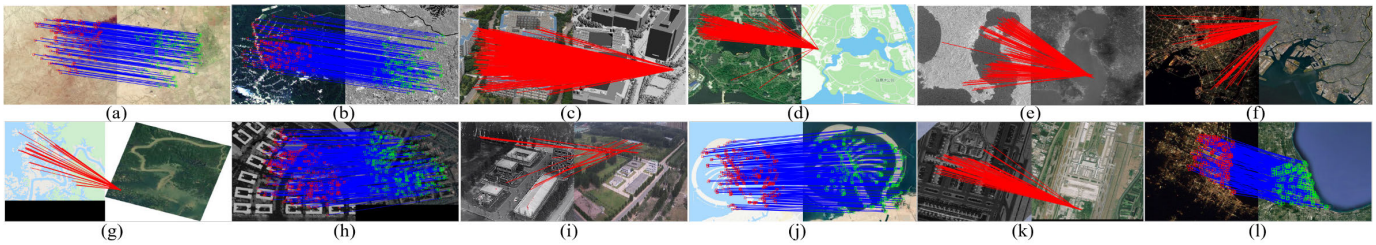


Fig. 10. Matching results of LNIFT on sample images. (a) Optical–optical. (b) Optical–infrared. (c) Optical–depth. (d) Optical–map. (e) Optical–SAR. (f) Day–night. (g) Rotation. (h) Optical–depth2. (i) Optical–infrared2. (j) Optical–map2. (k) Optical–SAR2. (l) Day–night2.

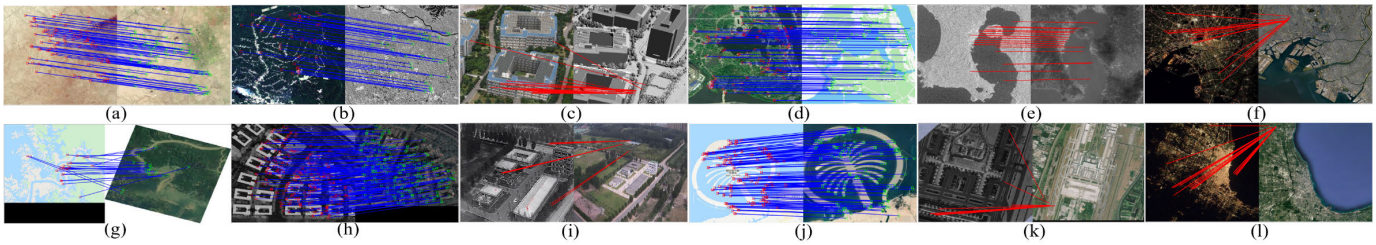


Fig. 11. Matching results of CoFSM on sample images. (a) Optical–optical. (b) Optical–infrared. (c) Optical–depth. (d) optical–map. (e) Optical–SAR. (f) Day–night. (g) Rotation. (h) Optical–depth2. (i) Optical–infrared2. (j) Optical–map2. (k) Optical–SAR2. (l) Day–night2.

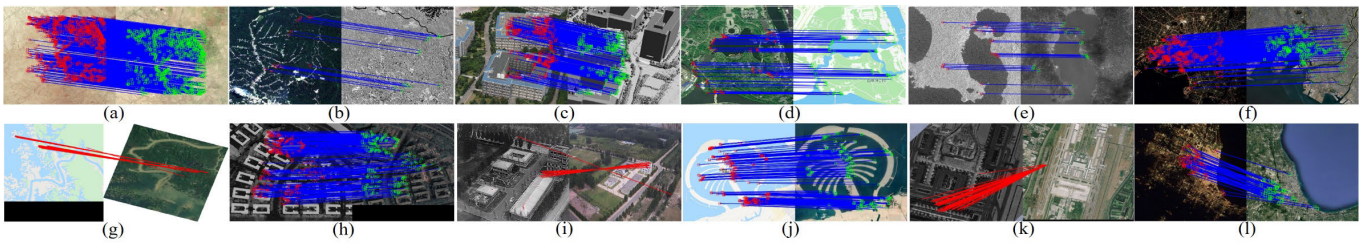


Fig. 12. Matching results of RIFT on sample images. (a) Optical–optical. (b) Optical–infrared. (c) Optical–depth. (d) optical–map. (e) Optical–SAR. (f) Day–night. (g) Rotation. (h) Optical–depth2. (i) Optical–infrared2. (j) Optical–map2. (k) Optical–SAR2. (l) Day–night2.

complexity. Therefore, the parameter N is set to 3 considering both SR and NCM, as well as the computational complexity.

2) Table III shows the relationship between SR and NCM and the radius R of the feature descriptors. When R is less than 48, the result of SR is poor. Different from

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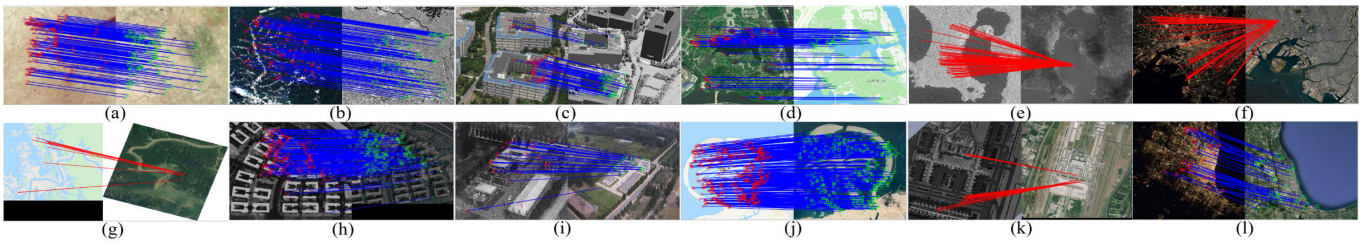


Fig. 13. Matching results of LPSO on sample images. (a) Optical–optical. (b) Optical–infrared. (c) Optical–depth. (d) Optical–map. (e) Optical–SAR. (f) Day–night. (g) Rotation. (h) Optical–depth2. (i) Optical–infrared2. (j) Optical–map2. (k) Optical–SAR2. (l) Day–night2.

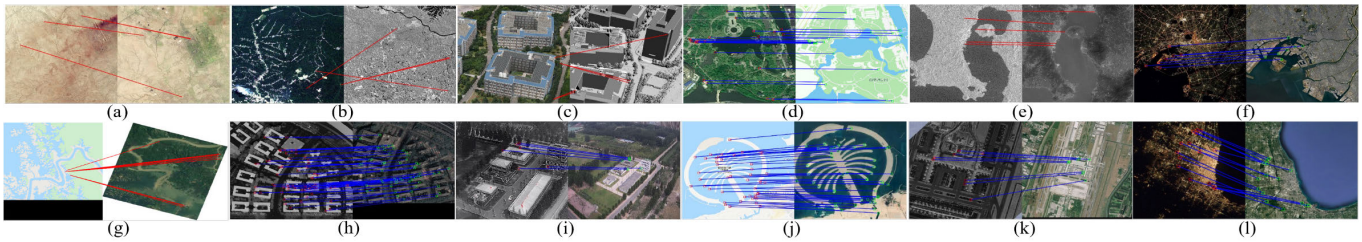


Fig. 14. Matching results of PSO-SIFT on sample images. (a) Optical–optical. (b) Optical–infrared. (c) Optical–depth. (d) Optical–map. (e) Optical–SAR. (f) Day–night. (g) Rotation. (h) Optical–depth2. (i) Optical–infrared2. (j) Optical–map2. (k) Optical–SAR2. (l) Day–night2.

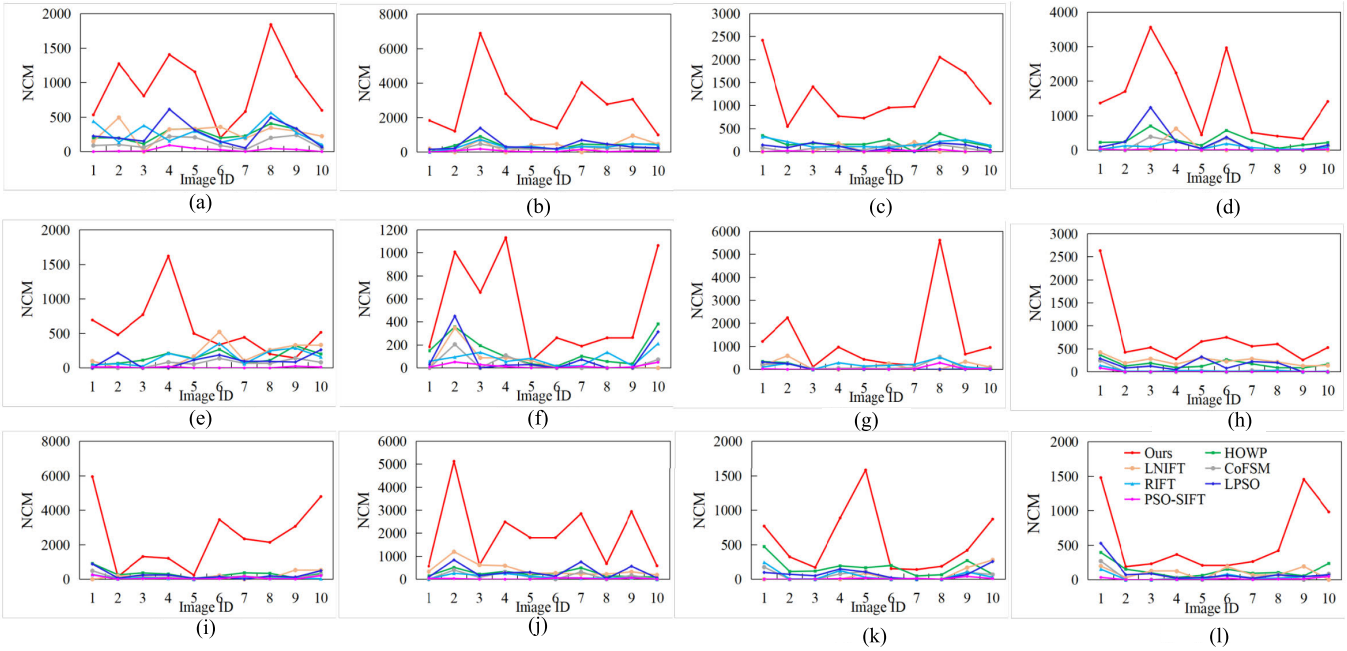


Fig. 15. Comparison of NCM metric on all types of images. (a)–(g) Results of NCM metric with five methods on Dataset 1. (h)–(l) Results of NCM metric with five methods on Dataset 2.

TABLE I

SETTINGS OF DIFFERENT PARAMETERS

experiments	variable	fixed Parameters
parameter N	$N = [1,2,3,4,5]$	$R = 48, R_f = 5$
parameter R	$R = [32,40,48,56,64]$	$N = 3, R_f = 5$
parameter R_f	$R_f = [4,5,6,7,8]$	$N = 3, R = 48$

TABLE II

RESULTS OF DIFFERENT PARAMETER N

Metric	$N, R = 48, R_f = 5$				
	1	2	3	4	5
NCM	181.417	415.667	595.583	687	819.833
SR/%	75	100	100	100	100

SR, as R increases, NCM decreases. When $R \geq 48$, the SR can reach 100%, and NCM is smaller than 500. To ensure the SR and take NCM into count, set R to 48.

3) For parameter R_f, the increase of R_f will improve the smoothness of the image, and the number of extracted feature points will decrease accordingly. When R_f is too large, the result of NCM will be poor, as shown in Table IV. In contrast, when R_f is too small, more noise will be retained, and the results of SR cannot be satisfactory. Considering the results of SR and NCM, we set the parameter R_f to 5 in this study.

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TABLE III
RESULTS OF DIFFERENT PARAMETER R

Metric	$R, N = 3, R_f = 5$				
	32	40	48	56	64
NCM	571.083	558	516.417	467.75	421.583
SR/%	91.667	91.667	100	100	100

TABLE IV
RESULTS OF DIFFERENT PARAMETER R_f

Metric	$R_f, N = 3, R = 48$				
	4	5	6	7	8
NCM	518.917	522.667	519.5	497.083	489
SR/%	100	100	100	100	100

D. Qualitative Evaluation

Twelve image pairs from two datasets are selected for qualitative comparison, as shown in Figs. 6 and 7. There exist severe image noises between the image pairs in Fig. 6(b) and (e). The day-night image pairs are actually not captured from the same objects since night images mainly capture nighttime light data, such as Figs. 6(f) and 7(e).

Image pairs in Fig. 6(a)–(f) and image pairs in Fig. 7(a)–(d) exist in different geometric variations, such as translation, rotation, and scale changes. In addition, there are significant NRDs between all test image pairs.

The results of the qualitative comparison are visualized in Figs. 8–14. As can be seen from Fig. 8, the proposed WSSF method matches well on all test image pairs, and the SR reaches 100%. Furthermore, the proposed method is superior to HOWP, PSO-SIFT, and RIFT in NCM metric.

Fig. 9 shows the matching results of HOWP. HOWP only fails to match the image pair in Fig. 9(g), and the SR reaches 91.667%. HOWP also achieves good results in NCM, except for the image pairs in Fig. 9(b) and (g), since HOWP uses phase features to calculate the main orientation of feature descriptors and limits the main orientation under 180°. Therefore, it is sensitive to huge rotation and could be affected by phase extreme value mutations and directional reversals.

Fig. 10 shows the matching results of LNIFT. LNIFT fails to match most of the image pairs and the SR only reaches 41.667%. LNIFT tries to extract the details of images for matching. However, this operation not only retains details, but also enhances image noises. In addition, when compared to the structure of images, details tend to be more unstable when multiple filtering is applied.

CoFSM succeeds in matching over half of the image pairs, and the SR reaches 58.333%. Fig. 11 shows the results of the qualitative evaluation with CoFSM. Since COF ignores those pixels located around a corner or in a long narrow strip area [57] and CoFSM does not consider the local details, CoFSM is not robust to MRSI matching. Thus, it is necessary to comprehensively consider the structure features and local details of images for MRSI matching.

RIFT matches most image pairs successfully, as shown in Fig. 12. The SR of RIFT is 75%. Different from other methods, RIFT constructs the MIM to build local descriptors, which consumes too much time. Therefore, RIFT discards the scale space and cannot achieve scale-invariance. Moreover, since MIM only has only six index values and lacks enough structure information, it might be greatly affected by image noise, as shown in Fig. 12(b).

Fig. 13 shows the matching results of LPSO on different types of images. LPSO fails to match in most cases, and the SR is 66.667%. Since LPSO constructs the scale space, so it can handle scale changes, as shown in Fig. 13(i) and (j). However, the matching results of LPSO are still poor. The main reason is GF cannot deal with significant NRD well. In addition, LPSO is very sensitive to huge rotation changes due to limits of the main orientation under 180° and affected by directional reversals.

The PSO-SIFT fails to match in most cases, and the SR only reaches 50%. In the image pair in Fig. 14(f), since the number of matching pairs is less than 10 and the fused image is not accurately aligned, so it is still considered a failed matching. Moreover, it can be seen from Fig. 14 that PSO-SIFT achieves the lowest NCM among all comparison methods.

E. Quantitative Evaluation

Fig. 15 shows the results of the quantitative evaluation of NCM across seven methods on two datasets (12 types in total). As can be seen from Fig. 15 and Tables V–VII, the proposed WSSF method successfully matches all image pairs, and NCMs are mostly more than 1000. The matching results of WSSF are robust because the WSSF highlights the structure features of images and weakens the influence of NRD. In addition, we also find that the matching results on optical-infrared types are much better than others, with the average NCMs (NCM_{ave}) exceeding 2000. The reason is mainly because the structure saliency of optical-infrared types is more significant than other image types.

HOWP possesses obvious advantages in six methods for comparison, and the overall SR reaches 94.167%, as shown in Table V. Since HOWP is very sensitive to rotation changes, it produces the worst performance on rotation image pairs, and the SR is only 50%. Different from the results on other image types, HOWP achieves much better results on optical-infrared image pairs, with the NCM_{ave} of 383.2 and 323.6, respectively. Comparing performance on optical-infrared images across seven methods, it can be inferred that better quality and stronger structure saliency of images will lead to more robust matching results.

As shown in Tables V–VII, the SR of LNIFT is 61.667% and the NCM_{ave} is 175.742. Since LNIFT pays more attention to details, it achieves relatively better performance on optical-infrared images among all datasets, with the average SR of two datasets is 70% and the NCM_{ave} is 263.5. In addition, it is worth noting that the SR of LNIFT on the first optical-map dataset is 10% and on the second is 100%. The performance of matching results on two optical-map datasets shows that the similarity scores of corresponding matching points can be low due to the details of different manifestations of MRSIs. Therefore, LNIFT is less robust for MRSI matching.

CoFSM mainly uses COF to retain structure features from images for matching. The SR of CoFSM is 70% and better

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TABLE V
RESULTS OF SEVEN METHODS IN TERMS OF FOUR EVALUATION METRICS

	WSSF	HOWP	LNIFT	CoFSM	RIFT	LPSO	PSO-SIFT
SR	**100%**	94.167%	61.667%	70%	93.636%	76.667%	55%
NCM	**1256.783**	211.183	175.742	100	141.127	180.808	26.558
RMSE	**1.926**	2.415	5.080	4.298	2.753	3.838	5.449
RCM	8.969%	**23.362%**	1.883%	3.383%	8.194%	5.108%	15.059%

TABLE VI
COMPARISONS ON SR METRIC OF ALL IMAGE TYPES

Method	SR/%					
	1	2	3	4	5	6
WSSF	**100**	**100**	**100**	**100**	**100**	**100**
HOWP	**100**	**100**	90	**100**	90	**100**
LNIFT	90	70	20	10	70	40
CoFSM	**100**	**100**	80	50	70	40
RIFT	**100**	**100**	**100**	**100**	**100**	**100**
LPSO	**100**	**100**	70	70	70	60
PSO-SIFT	60	90	40	40	50	70
Method	SR/%					
	7	8	9	10	11	12
WSSF	**100**	**100**	**100**	**100**	**100**	**100**
HOWP	50	**100**	**100**	**100**	**100**	**100**
LNIFT	60	**100**	70	**100**	30	80
CoFSM	**100**	40	90	60	50	60
RIFT	90	80	90	90	50	80
LPSO	20	70	90	**100**	80	90
PSO-SIFT	70	10	80	50	50	50

Type 1-Type 7 belong to Dataset 1; Type 8- Type 12 belong to Dataset 2. 1. optical-optical; 2. optical-Infrared; 3. optical-depth; 4. optical-map; 5. optical-SAR; 6. day-night; 7. Rotation; 8. optical-depth; 9. optical-Infrared; 10. optical-map; 11. optical-SAR; 12. day-night.

TABLE VII
COMPARISONS ON NCM_{ave} METRIC OF ALL IMAGE TYPES

Method	NCM_{ave}					
	1	2	3	4	5	6
WSSF	**949.3**	**2752.7**	**1263.6**	**1494.5**	**571.2**	**508.8**
HOWP	241	383.2	197.5	287.7	155.1	142.6
LNIFT	271.7	350.8	39.1	62.5	181.8	58.7
CoFSM	128.6	209.4	73.1	115.7	63.2	40.8
RIFT	269.7	325.2	172.1	95	158.2	82.9
LPSO	259.3	426.9	100	239	106	92
PSO-SIFT	25.2	69.3	11	11.8	8.5	17.2
Method	NCM_{ave}					
	7	8	9	10	11	12
WSSF	**1267.1**	**723.3**	**2469.7**	**1949.2**	**551.3**	**580.7**
HOWP	71.8	168	323.6	252	172.5	139.2
LNIFT	155.8	236.4	176.1	422.2	53.6	100.2
CoFSM	139.3	32.5	150.8	138	53.5	55.1
RIFT	186.1	30.5	85.8	105.7	54	41.2
LPSO	57	137.4	246.9	321.2	82.6	101.4
PSO-SIFT	41.3	8.3	80.3	24.1	9.1	12.6

Type 1-Type 7 belong to Dataset 1; Type 8- Type 12 belong to Dataset 2. 1. optical-optical; 2. optical-Infrared; 3. optical-depth; 4. optical-map; 5. optical-SAR; 6. day-night; 7. Rotation; 8. optical-depth; 9. optical-Infrared; 10. optical-map; 11. optical-SAR; 12. day-night.

than LNIFT. The results presented in Tables V–VII reveal that relying solely on image structure features for matching is insufficient. To achieve robust matching performance, it is necessary to comprehensively consider the structure features and local details of images.

RIFT successfully realizes matching in most images, with the performance only decreasing by 0.531% compared to HOWP. The SR of RIFT reaches 100% on Type 1–Type 6 image pairs, as shown in Table VI. However, it is difficult to obtain robust results on Dataset 2, especially in the

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TABLE VIII
COMPARISONS ON RMSE METRIC OF ALL IMAGE TYPES

Method	RMSE					
	1	2	3	4	5	6
WSSF	**1.859**	**1.876**	**1.940**	**1.870**	**2.000**	1.878
HOWP	1.892	2.185	2.745	1.898	2.766	**1.855**
LNIFT	2.800	4.427	8.415	9.198	4.435	6.806
CoFSM	1.861	1.901	3.549	5.907	4.404	6.714
RIFT	1.891	2.180	1.942	2.106	2.418	2.345
LPSO	2.080	1.959	4.463	4.295	4.403	5.359
PSO-SIFT	5.003	2.412	6.786	6.558	5.850	4.632
Method	RMSE					
	7	8	9	10	11	12
WSSF	1.896	**1.964**	**1.867**	1.888	2.012	2.061
HOWP	6.087	1.983	1.906	**1.779**	**1.950**	**1.939**
LNIFT	5.214	2.060	4.414	2.028	7.605	3.564
CoFSM	**1.730**	6.719	2.644	5.008	6.020	5.123
RIFT	2.706	4.429	3.152	3.016	6.207	4.102
LPSO	8.363	3.998	2.671	1.952	4.104	2.415
PSO-SIFT	4.206	9.193	3.205	5.726	5.958	5.858

Type 1-Type 7 belong to Dataset 1; Type 8- Type 12 belong to Dataset 2. 1. optical-optical; 2. optical-Infrared; 3. optical-depth; 4. optical-map; 5. optical-SAR; 6. day-night; 7. Rotation; 8. optical-depth; 9. optical-Infrared; 10. optical-map; 11. optical-SAR; 12. day-night.

TABLE IX
RESULTS OF ABLATION EXPERIMENT ABOUT PSATF SCALE SPACE IN TERMS OF FOUR EVALUATION METRICS

Method	NCM	RMSE	RCM	SR
1	**1036.557**	**1.943**	**8.370**	**100**
2	661.557	2.514	5.649	94.286
3	519.729	2.370	5.712	94.286
4	618.271	2.310	5.038	95.714

1.PSATF; 2. GF; 3. LNF; 4. CoF.

TABLE X
RESULTS OF ABLATION EXPERIMENT ABOUT $G^{\theta}_{2,\sigma}-$SSM IN TERMS OF FOUR EVALUATION METRICS

Method	NCM	RMSE	RCM	SR
1	532.386	3.774	5.048	80
2	66.529	8.359	0.497	20
3	961.357	2.158	7.446	97.143
4	1002.186	2.037	7.527	98.571
5	605.086	3.605	4.747	82.857
6	239.829	5.938	1.915	48.571
7	975.371	2.194	7.759	98.571
8	**1036.557**	**1.943**	**8.370**	**100**

1. M_{max}; 2. M_{min}; 3. ECM; 4. SSM; 5. $M_{max}+G^{\theta}_{2,\sigma}$; 6. $M_{min}+G^{\theta}_{2,\sigma}$; 7. ECM + $G^{\theta}_{2,\sigma}$; 8. $G^{\theta}_{2,\sigma}-$SSM.

optical-depth, optical-SAR, and day-night image pairs. The NCM_{ave} of these types are only 30.5, 54, and 41.2. The RIFT achieves relatively better results in the optical-map images, where the SR is 90% and the NCM_{ave} is 105.7. This is because most of the image pairs of Dataset 2 have apparent scale differences.

Therefore, since RIFT discards the scale space to reduce computational complexity, it is not robust to scale changes.

The overall SR of LPSO is 76.667% and the NCM_{ave} is 180.808. Tables VI and VII list the results of SR and NCM_{ave}, respectively. As seen, LPSO performs relatively worse results on two dataset images. The SR of LPSO on rotation images is only 20%, and the NCM_{ave} is 57. Similar to HOWP, LPSO also uses phase features to calculate the descriptors. However, the matching results of LPSO are much worse than HOWP, the reason mainly because the LPSO does not notice the importance of edge-preserve filtering.

PSO-SIFT achieves the worst performance among all compared methods, with SR only of 55% on two datasets. In most cases, the NCM of PSO-SIFT is less than 20. Although PSO-SIFT uses the second-order Sobel operator to redefine the traditional image gradient, the results are still poor.

Table VIII summarizes the RMSE of the seven comparison methods on each type; for a failed matching trial, the RMSE is set to 10 pixels. It can be found that our method has the highest average matching accuracy among all image pairs. The matching accuracy of the proposed method is slightly inferior to HOWP on some image pairs. Compared with HOWP, the matching accuracy of WSSF is reduced by 0.023, 0.109, 0.062, and 0.122 pixels, respectively. Combining all types of MRSI datasets, the RMSE of WSSF is 1.926 pixels, and the NCM_{ave} is 1256.783.

HOWP has a good matching performance on all types of image pairs except rotation types, especially on the day-night, optical-map, and optical-SAR types of Dataset 2, with RMSE of 1.939, 1.950, and 1.779 pixels, respectively. For rotation image pairs, the SR of HOWP only has 50%, and the RMSE reaches 6.087 pixels.

LNIFT achieves poor results in experiments. The matching error of LNIFT on 12 types of image pairs reaches 5.080 pixels. Due to imaging differences of multimodal images, their details will vary greatly. Thus, LNIFT performs poorly on most image pairs, especially on the optical-map type of Dataset 1. Although CoFSM uses COF to retain the structure features of images, the RMSE of CoFSM also reaches 4.298 pixels. COF ignores those pixels located around corners, and CoFSM redefines the image gradient through the second-order Sobel operator. These operations will lose a lot of useful information.

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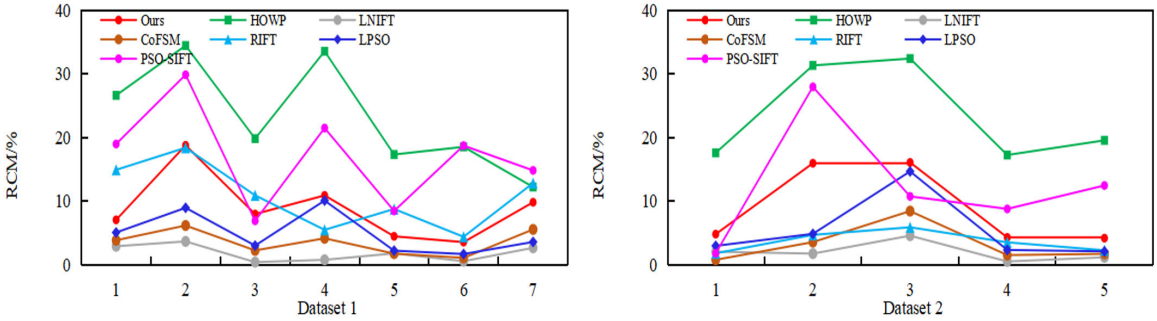


Fig. 16. Comparison of RCM metric on two datasets.

Compared with the performance on Dataset 2, RIFT has a relatively good matching accuracy on image pairs of Dataset 1, and the RMSE is about 2.227 pixels. However, the RIFT's average matching error reaches 4.1812 pixels on Dataset 2, with the RMSE of 4.429, 3.152, 3.016, 6.207, and 4.102 pixels, respectively. In addition, the NCMs of RIFT on Dataset 2 are also poor. The NCM_{ave} of optical-SAR, optical-depth, and day-night images are only 54, 30.5, and 41.2, respectively, which is far inferior to WSSF and HOWP. The reason is that the RIFT method is free of the construction of scale space, so it is more sensitive to the matching of scale images.

Since LPSO does not filter the matched image, image noise greatly affects the matching accuracy of LPSO, with the RMSE being 3.838 pixels among all image pairs. As seen, LPSO performs worst on rotation images, the RMSE reaches 8.363 pixels, and the NCM_{ave} only has 57. This result is similar to HOWP, which also performs the worst on rotation images. Different from the other six types of images, LPSO performs better on optical-optical images because there are small NRD between images.

PSO-SIFT has the worst matching accuracy among all comparison methods, and the RMSE of PSO-SIFT on 12 types of image pairs is 5.449 pixels, with the SR is 55%. Although PSO-SIFT redefines the image gradient through the second-order Sobel operator, the results are still poor. The reason is PSO-SIFT should pay more attention to structure features instead of texture features.

Fig. 16 and Table V show the RCM of seven comparison methods. The RCM of HOWP is the highest among all methods, which reaches 23.362% on two datasets. For feature point extraction, HOWP proposes the aggregation feature strategy to extract blob points and corner points [24]. In contrast, LNIFT has the lowest RCM within the seven methods, and the RCM of the two datasets is 1.883%. WSSF and RIFT have RCM scores of 8.969% and 8.194%, respectively. Since we do not apply additional constraints when using the FAST detector for feature extraction, our RCM score is not satisfactory and the running time for one image pair reaches about 24 s.

F. Ablation Study on Scale Space

To evaluate the effectiveness of the proposed PSATF, we randomly choose 70 image pairs from two datasets, and the selected image pairs covered all image types. We replace the PSATF scale space with the construction from GF, locally normalization filtering (LNF) [1], and COF [11], and keep other parts unchanged. Table IX shows the comparison results in terms of NCM, RCM, SR, and RMSE.

Since PSATF uses the entire structure information to guide bilateral filtering, it can better preserve edges than other filters and facilitate improving the accuracy of matching. The SR of PSATF is 100%, and through using PSATF, our method achieves the best matching results among all metrics.

GF achieves the worst RMSE results. Since GF is not designed for edge preserving, it is not adaptable for MRSI matching. To enhance image details, LNF uses the input image subtracting the result of mean filtering. This operation not only enhances detailed information, but also increases the noise. Therefore, it can be found that the results of LNF are not satisfactory. The SR of LNF is 94.286% and the RCM is only 5.712%.

COF and PSATF both belong to joint bilateral filtering. Different from PSATF, COF ignores those pixels located around a corner or in a long narrow strip area, and the SR of COF is 95.714%. Although COF does not perform the best among all compared filters, it still achieves better results than GF and LNF. Therefore, it is necessary to explore the influence of image edges on MRSI matching.

G. Ablation Study on $G^{\theta}_{2,\sigma}$ − SSM

Designing a robust radiation-invariant feature is necessary for MRSI matching. We conduct an ablation experiment on 70 randomly selected image pairs to demonstrate the effectiveness of $G^{\theta}_{2,\sigma}$−SSM. We replace $G^{\theta}_{2,\sigma}$−SSM with $M_{\max}$, $M_{\min}$, ECM, SSM, $M_{\max}+G^{\theta}_{2,\sigma}$, $M_{\min}+G^{\theta}_{2,\sigma}$, ECM+$G^{\theta}_{2,\sigma}$, and $G^{\theta}_{2,\sigma}$−SSM, respectively. Table X summarizes the comparison results in terms of NCM, RCM, SR, and RMSE.

From Table X, it can be found that the proposed $G^{\theta}_{2,\sigma}$−SSM achieves the best results on all comparison metrics. In addition, the ECM feature proposed in this article is more robust compared with the phase features. The SR of ECM, SSM, and ECM+$G^{\theta}_{2,\sigma}$ are 97.143%, 98.571%, and 98.571%, respectively. Therefore, the ECM feature proposed in this article is an important guarantee for successful matching.

The matching accuracy of all methods is improved by using $G^{\theta}_{2,\sigma}$, and the final $G^{\theta}_{2,\sigma}$−SSM achieves the best matching performance. For example, the SR of $M_{\max}+G^{\theta}_{2,\sigma}$ is 2.857% higher than $M_{\max}$ through using $G^{\theta}_{2,\sigma}$ since steerable filtering can provide more details by collecting more information from different angles. Therefore, in order to ensure the performance

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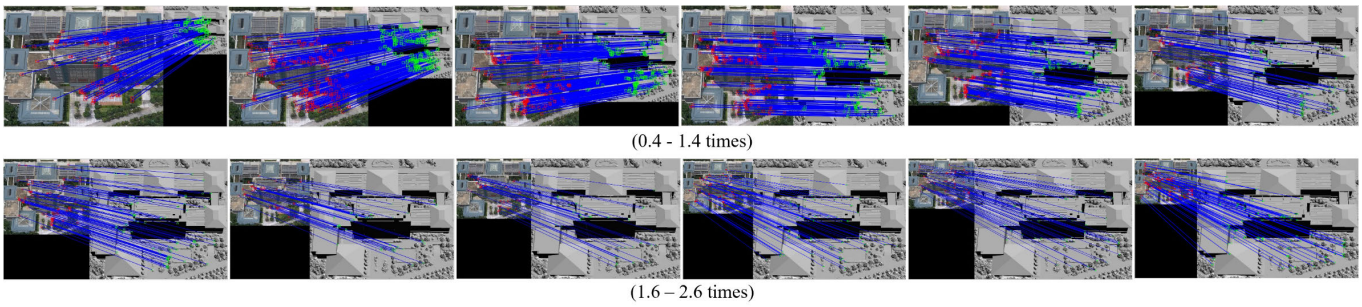


Fig. 17. Matching results under different scale differences of the WSSF method.

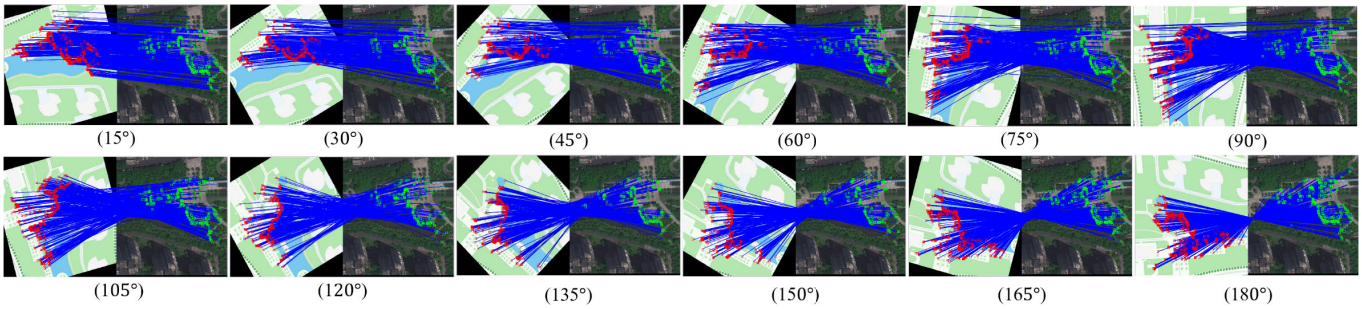


Fig. 18. Matching results under different rotation differences of the WSSF method.

of the proposed WSSF method, it is necessary to adopt $G^{\theta}_{2,\sigma}$ to further enhance the SSM feature.

H. Scale Invariance Analysis

In order to verify the scale invariance of WSSF, a pair of optical-depth images is selected for testing, as shown in Fig. 17. We generate a total of 12 scale images (0.4–2.6 times) by spacing the depth images by 0.2 times. Fig. 17 shows the matching results across 12 groups of scale images. It can be found that the proposed WSSF method can complete the matching task well on all 12 groups. Even with a scale difference of 0.4 and 2.6 times, there are still enough correct matching pairs. Although the number of correct matching pairs obtained by WSSF gradually decreases with the change of scale difference, the NCMs are always more than 50.

Since the structural saliency of the image is considered as the basis of the constructed scale space, and $G^{\theta}_{2,\sigma}$ − SSM is adopted to calculate the robust descriptors, the proposed method can deal with the scale changes well.

I. Rotation Invariance Analysis

Matching results of rotation images are very important. In this section, we evaluate the rotation invariance of WSSF. We take the optical image as the reference image and generate a total of 12 sensed images by rotating the map images clockwise at intervals of 15°, ranging from 15° to 180°. Experimental images are presented in Fig. 18.

Fig. 18 shows the matching results in 12 sets of rotation images. From Fig. 18, we can find that no matter how large the rotation, WSSF completes the matching task well and the number of correct matching pairs is considerable. Even rotation of 180°, the NCMs are far more than 300, and the matching pairs are still distributed uniformly. Therefore, our method can effectively handle the rotation changes of MRSI.

V. Conclusion

This article develops a multimodal feature matching method, called WSSF, and the WSSF is adopted to deal with the problem of NRD. By constructing the edge-preserving scale space, structure features can be retained, improving the similarity of images. Furthermore, the ECM, phase features, and steerable filtering are combined to construct the $G^{\theta}_{2,\sigma}$−SSM, which is instead of the traditional image gradient map. The reliability of the proposed WSSF method is evaluated on two MRSI datasets, which contain 120 image pairs. The experimental results validate that the proposed WSSF outperforms other compared methods. In addition, the robustness of WSSF to scale and rotation changes is also verified.

Although WSSF achieves satisfactory matching accuracy in experiments, it is relatively sensitive to weak structure images, in which the discontinuity issue of edge structure exists or the structural information of image is not apparent. In addition, WSSF consumes too much time. Accordingly, future work will focus on improving the robustness of weak structure images by collecting aggregation features and reducing the computational complexity by optimizing the feature extraction.

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